FT1.1 — Mathematical Model for Sudoku / Generalized Sudoku

Sistema Operador Unificador Fenomenológico General

Abstract

This document defines a mathematical formulation for solving Sudoku and generalized Sudoku boards by rep

Equation of Solution

$$S^* = \mathrm{argmin}_S \left[\Phi(S) + \lambda \sum_{x,y} |z_{k+1} - z_k| \right]$$

Condition: $\Phi(S) = 0$

Variables

$S(x,y,z)$: board state
S^* : optimal solution
$\Phi(S)$: constraint violation function
λ : depth regulation factor
x,y : board coordinates
z : logical exploration depth
$\beta \in \{-1,1\}$: depth displacement bias

Domain

$D = \{1,2,...,n^2\}$

Classic Sudoku:
Board: 9x9
Subgrid: 3x3
Domain: {1..9}

Interpretation

The Sudoku board is embedded into a 3D configuration space.
XY preserves the geometric board structure.
Z allows exploration of alternate configurations without classical backtracking.
A valid solution is achieved when $\Phi(S)=0$.

Conceptual Flow

Sudoku Board
↓
Cartesian Representation (x,y,z)
↓
Constraint Evaluation Φ(S)

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   ↓
Depth Exploration
   ↓
Conflict Minimization
   ↓
Solution S*
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